# Phase portraits for equal real Eigenvalues

Now we want to look at the phase portraits of systems with equal real Eigenvalues.

We'll first consider what happens when the coefficient matrix $A$ is nonsingular, then we'll look later at what happens when it's singular.

## The nonsingular matrix

If the coefficient matrix $A$ is **nonsingular**, which means its determinant is non-zero $|A| \neq 0$ and $0$ is not an Eigenvalue of $A$, then there are two cases:

1. The repeated Eigenvalue produces multiple linearly independent Eigenvectors
2. The repeated Eigenvalue produces only one Eigenvector

In the first case, where the repeated Eigenvalue produces multiple linearly independent Eigenvectors, here's what we need to know:

1. Any vector in the plane is a linear combination of two different linearly independent Eigenvectors of the matrix $A$. Therefore, all vectors in the plane are Eigenvectors of $A$ associated with our Eigenvalue, so all trajectories are lines passing through the origin.
2. The direction along the linear trajectories depends on the sign of the associated Eigenvalue. A linear trajectory associated with a positive Eigenvalue will always move away from the origin, while a

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linear trajectory associated with a negative Eigenvalue will always move toward the origin.

3. The equilibrium of a positive Eigenvalue is an unstable singular node that repels all trajectories, while the equilibrium of a negative Eigenvalue is an asymptotically stable singular node that attracts all trajectories.

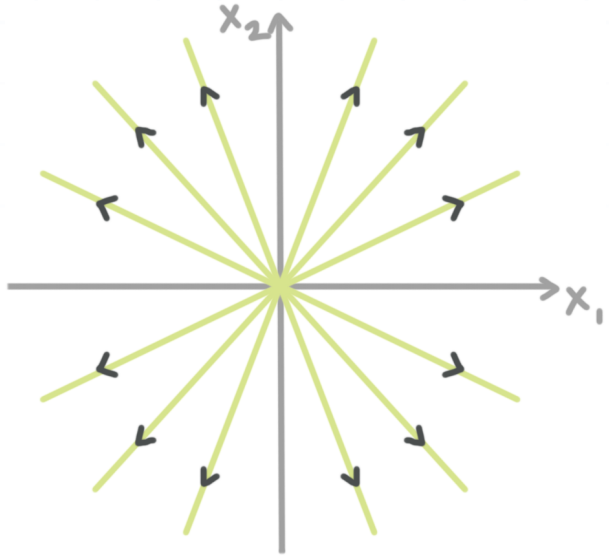
4. To check for the direction of the trajectories for any case of equal real Eigenvalues, we can use the $(1,0)$ test, which means we'll look at the direction at the point $(1,0)$.

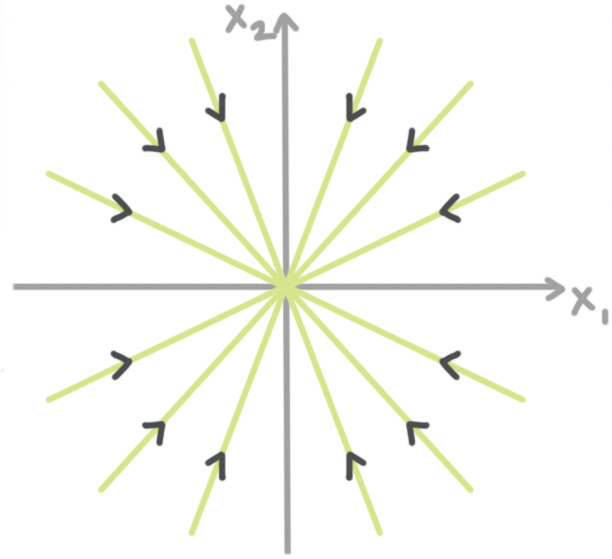
Below is a table that summarizes information about the phase portrait, based on the sign of the equal real Eigenvalues.

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## EQUAL REAL EIGENVALUES, TWO EIGENVECTORS

| | **Positive** | **Negative** |
|---|---|---|
| | $\lambda_1 = \lambda_2 > 0$ | $\lambda_1 = \lambda_2 < 0$ |
| **Equilibrium** | Singular Node | Singular Node |
| **Stability** | Unstable | Asymptotically stable |
| | (Repeller) | (Attractor) |
| **Direction** | $(1,0)$ | $(1,0)$ |
| **Sketches** | | |




In the second case, where the repeated Eigenvalue produces only one Eigenvector, here's what we need to know:

1. The single straight line corresponds to the one Eigenvector of the system. An Eigenvector $\overrightarrow{k_1} = (k_1, k_2)$ will lie along the line $y = (k_2/k_1)x$.

2. The direction along the linear trajectory is dependent on the sign of the associated Eigenvalue. A linear trajectory associated with a

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positive Eigenvalue will always move away from the origin, while a linear trajectory associated with a negative Eigenvalue will always move toward the origin.

3. The equilibrium of a positive Eigenvalue is an unstable node that repels all trajectories, while the equilibrium of a negative Eigenvalue is an asymptotically stable node that attracts all trajectories.

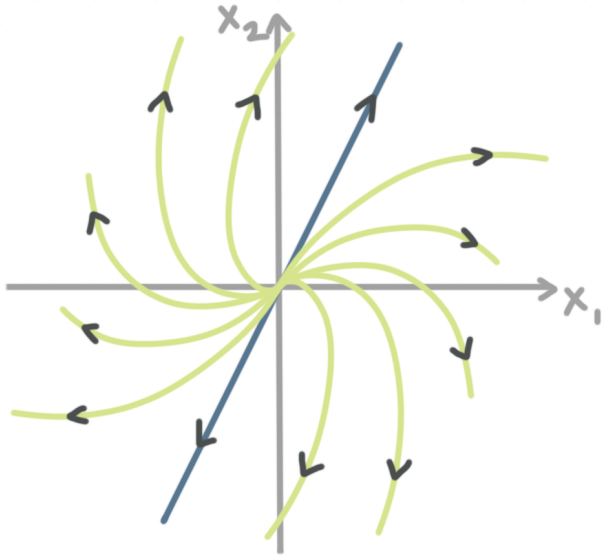
4. To check for the direction of the trajectories for any case of equal real Eigenvalues, we can use the $(1,0)$ test, which means we'll look at the direction at the point $(1,0)$.

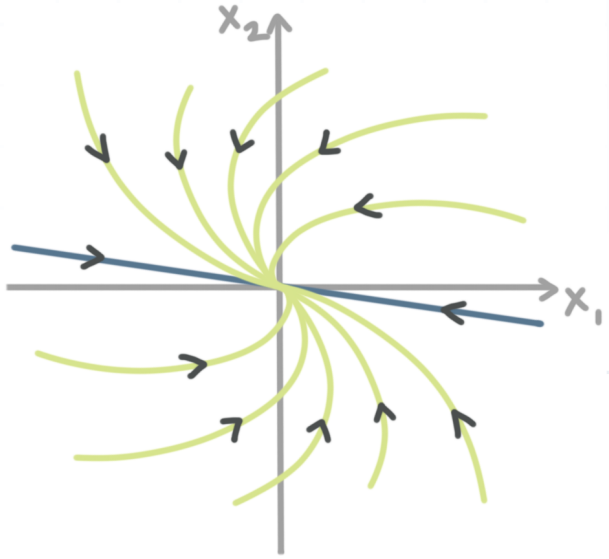
Below is a table that summarizes information about the phase portrait, based on the sign of the equal real Eigenvalues.

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## EQUAL REAL EIGENVALUES, ONE EIGENVECTOR

| | Positive | Negative |
|---|---|---|
| | $\lambda_1 = \lambda_2 > 0$ | $\lambda_1 = \lambda_2 < 0$ |
| **Equilibrium** | Node | Node |
| **Stability** | Unstable | Asymptotically stable |
| | (Repeller) | (Attractor) |
| **Direction** | $(1,0)$ | $(1,0)$ |
| **Sketches** | | |




Let’s do an example so that we can see how to build a phase portrait for equal real Eigenvalues. We’ll look at a nonsingular coefficient matrix, for which only one Eigenvector is produced from the equal real Eigenvalues.

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**Example**

Sketch the phase portrait of the system.

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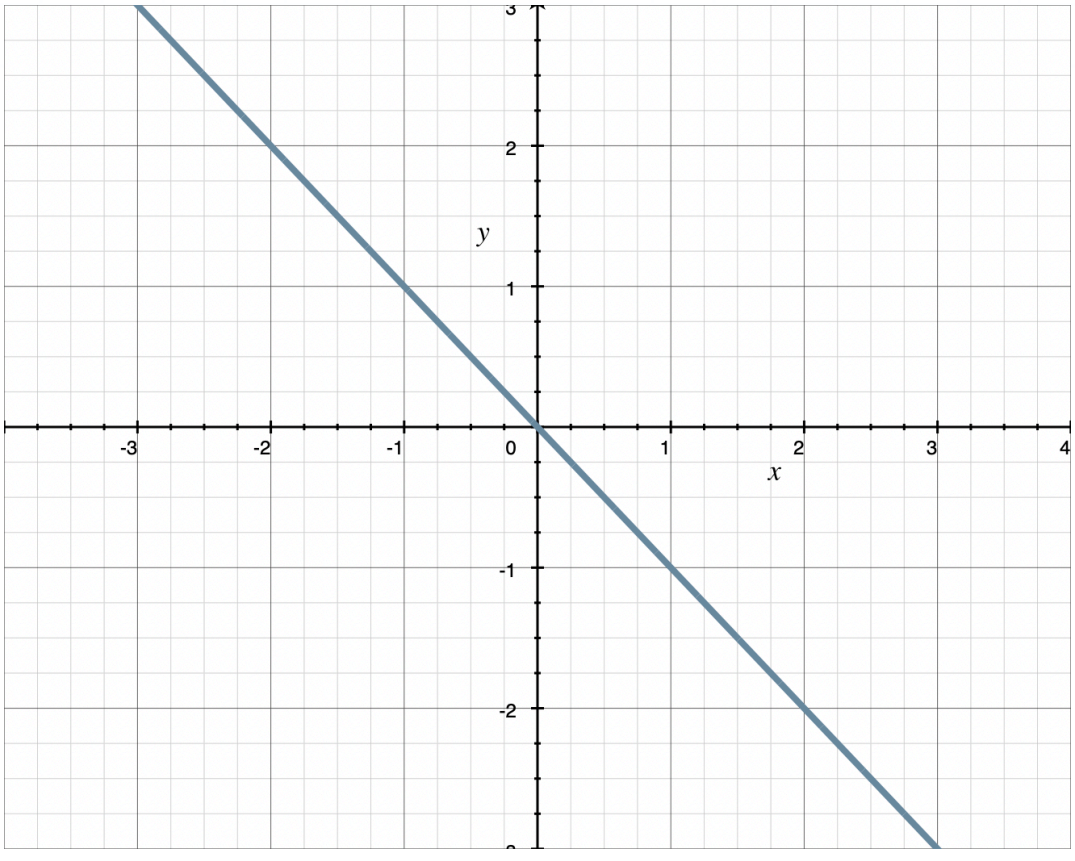
$$x_1'(t) = 2x_1 - x_2$$

$$x_2'(t) = x_1 + 4x_2$$

This is one of the examples we looked at when we learned to solve homogeneous systems with equal real Eigenvalues, and we found that the Eigenvalues for the system were $\lambda_1 = \lambda_2 = 3$. The associated single Eigenvector was

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

The Eigenvector $\overrightarrow{k_1} = (1, -1)$ lies along the line $y = -x$, so we'll sketch this line.




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The Eigenvalue associated with $\overrightarrow{k_1} = (1, -1)$ is $\lambda = 3$, which means the direction along that trajectory is away from the origin. Therefore, we're dealing with an unstable node that repels all trajectories.

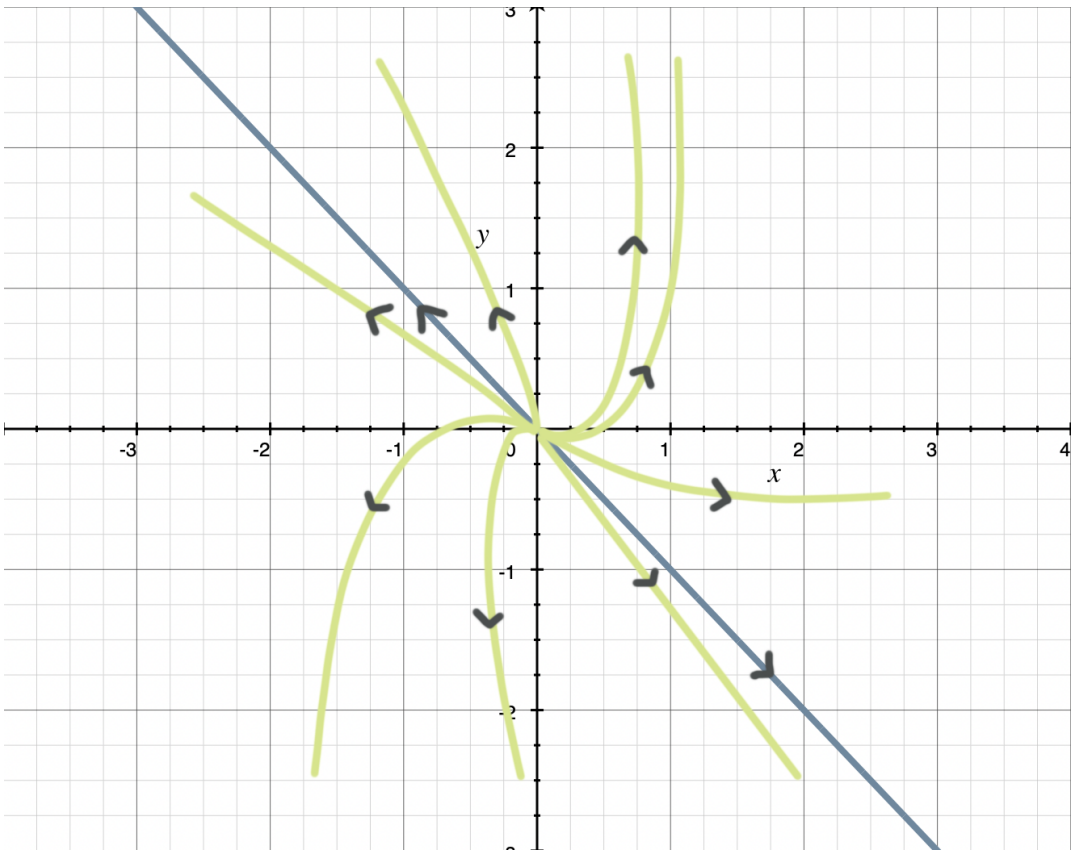
If we test the vector $\overrightarrow{x} = (1,0)$ in the matrix equation associated with the system, we get

$$\overrightarrow{x}' = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix} \overrightarrow{x}$$

$$\overrightarrow{x}' = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\overrightarrow{x}' = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(2,1)$ (into the first quadrant), which means that the phase portrait must look something like



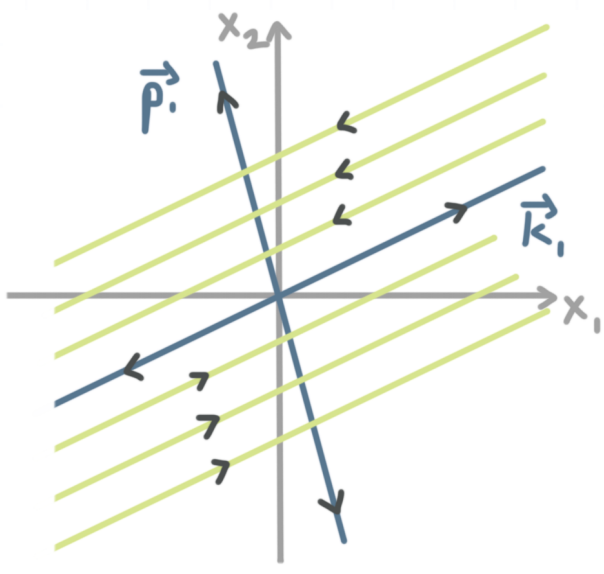

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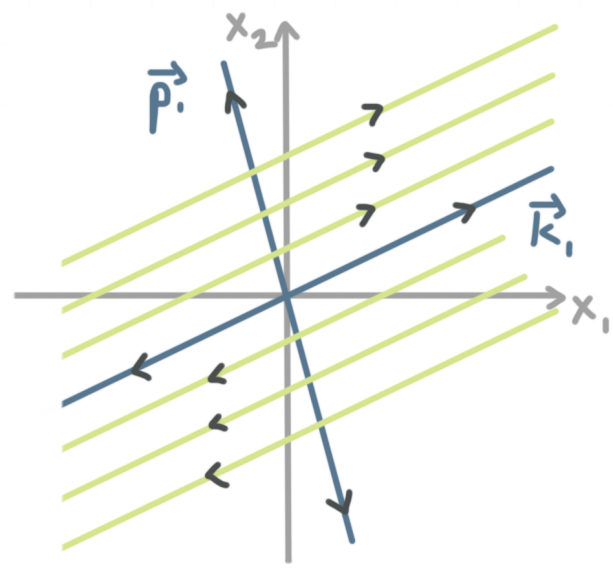
## The singular matrix

If the coefficient matrix $A$ is **singular**, which means its determinant is $0$, $|A| = 0$, then its Eigenvalues are both $\lambda_1 = \lambda_2 = 0$, and there are two cases:

1. Every entry in the coefficient matrix $A$ is $0$ ($A$ is the zero matrix $\vec{O}$), in which case every point in the plane is an equilibrium point.

2. The Eigenvalue $\lambda_1 = \lambda_2 = 0$ produces one Eigenvector $\vec{k_1}$, in which case we need to find the vector $\vec{p_1}$ using $(A - \lambda I)\vec{p_1} = \vec{k_1}$. The trajectories are parallel to the Eigenvector $\vec{k_1}$, and every point on $\vec{p_1}$ is an equilibrium point.




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